

Project Design Document

What problem are we solving?

Review the current SLAs for each public service Urban Forestry provides and recommend practical key performance indicators (KPIs) to track compliance with current SLAs.

What is the proposed solution?

Perform an analysis of all tree maintenance service requests for the Houston Parks & Recreation Department's Urban Forestry Division - from 2020 to 2025 - to answer the following business-related questions:

- At what time of the day are most service requests received?
- What percentage of requests failed their SLA?
- Are there any patterns or anomalies in request volume each year?
- What are practical key performance indicators (KPIs)?

What is the project scope?

1. Gather raw data from 2020 to 2025 from the following sources:
 - Retrieve service request records from the City of Houston's 311 Help Center
 - Retrieve service request records from the Houston Parks & Recreation Department
2. Import raw data into Jupyter Notebook.
3. Clean each table to address missing values and to standardize column names.
4. Import cleaned tables into database/SQL tool (DBeaver 26.1.5).
5. Create a staging table – combine all tables from each year.
6. Create dimension tables for columns with repeating values.
7. Create a fact table to store primary keys relating to each dimension table.
8. Import fact table and all dimension tables into Tableau Public.
9. Build star schema to link dimension tables to the fact table.
10. Perform analysis to address questions.
 - **Service Requests Received by Hour:** Measure the total number of service requests received each hour for 24 hours over the 5-year timeframe. Use Created On.
 - Display with vertical bar chart

- **Service Requests Received Per Month:** Measure the total number of service requests received each month to determine any cyclical or seasonal patterns. Use factd365srs and Date_Key (month).
 - Display with horizontal line chart
- **Average Service Time:** Measure the average number of days it takes to complete each service request by service type. Use Service Time and Service Type.
 - Display with horizontal line chart, grouped by Service Type
- **SLA Success Rate:** Calculate the percentage of service requests that were completed before the SLA. Use Created On, Finish By, and Actual Finish.
 - Display with vertical bar chart

What is the dashboard design?

The dashboard design will show the 4 charts mentioned in line 10 of the project scope. Display with a single page quad design.

Each chart will serve as historical data to support the new SLA recommendations.

Set each chart to be used as a filter.